Jiahao Zhang

Machine Learning PhD Student
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Google Scholar | GitHub | LinkedIn

Research Interests

Machine Learning, Large Language Models & Foundation Models, Interpretable AI & Alignment, Mechanistic Interpretability, Foundation Models for Life Sciences.

Education

Mohamed bin Zayed University of Artificial Intelligence — Aug. 2025 – Present
Ph.D. in Machine Learning — Abu Dhabi, UAE
Supervisor: Prof. Lijie Hu; Secondary supervisor: Prof. Kun Zhang.

Huazhong University of Science and Technology — Sept. 2021 – Jun. 2025
B.Eng. in Electronic Information — Wuhan, China

University of California, Berkeley — Jan. 2024 – May 2024
Visiting Student — Berkeley, CA, USA

Publications

* Equal contribution (co-first authors); † corresponding author.

Accepted

[1] Xijie Gong*, Tingxu Han*, **Jiahao Zhang**, Wei Song, Ziqi Ding, Hanqi Yan, Youcheng Sun, Lijie Hu. *How Do Agentic LLMs Decide to Call Tools? A Scaffold Default Controlled by Suppression.* NeurIPS 2026. [arXiv] [Code]

[2] Haodong Wu, **Jiahao Zhang**, Lijie Hu, Yongqi Zhang†. *From Instance Selection to Fixed-Pool Data Recipe Search for Supervised Fine-Tuning.* EMNLP 2026, Main Conference. [arXiv] [Code]

[3] Peng Cui*, **Jiahao Zhang***, Lijie Hu†. *Bayesian Gated Non-Negative Contrastive Learning.* ICML 2026. [arXiv] [Code]

[4] **Jiahao Zhang***, Zeqing Zhang*, Di Wang, Lijie Hu†. *Controlling Repetition in Protein Language Models.* ICLR 2026, Poster. [arXiv] [Code]

Preprint

[5] Xinyue Xu*, **Jiahao Zhang***, Lijie Hu†, Peter Hase, Hao Wang†. *D-Scope: Decomposing and Steering Diffusion Transformers with Sparse Autoencoders.* arXiv:2609.39625, 2026. [arXiv] [Demo]

Publications (continued)

Under review

[6] **Jiahao Zhang**[*], Yueying Hu[*], Xinyue Xu, Lijie Hu[†]. *Rethinking Circuit Discovery with Learnable Corruptions.* Submitted to ICLR 2027.

[7] **Jiahao Zhang**, Wenshuo Dong, Jie Li, Wei Tang, Lijie Hu[†]. *Coarse-to-Fine Neuron-Path Discovery for Interpretable Vision Transformers.* Submitted to ICLR 2027.

[8] Honglang Lu, **Jiahao Zhang**, Xingxiang Li, Jing Li, Yishuo Huang, Xiaoqin Zhong, Lijie Hu, Jun Ma, Zongyao Zhou[†]. *Solar-driven evapofiltration enables co-production of lithium and freshwater from extreme brines.* Submitted to Nature Sustainability, 2026. [Code]

Research Experience

Research Assistant, Westlake University — Jan. 2025 – Jul. 2025
Laboratory of Cell Ethology (CIS) — Hangzhou, China

Research Assistant, Westlake University — Jun. 2024 – Dec. 2024
Representation Learning Lab — Hangzhou, China

Data Science Research Intern, Grapedata (UC Berkeley) — Jan. 2024 – May 2024
Berkeley, CA, USA

Research Assistant, University of Wisconsin-Madison — Oct. 2023 – Dec. 2023
AI Lab (Chaowei Xiao) — Remote

Teaching

Teaching Assistant, MBZUAI — Fall 2026
Generative AI Memorization

Academic Service

Conference reviewer: ICML 2026; NeurIPS 2026; AAAI 2027; ICLR 2027.

Journal reviewer: IEEE Computational Intelligence Magazine; Neurocomputing; ACM Transactions on Probabilistic Machine Learning.

Honors and Awards

MBZUAI Conference Travel Grant	Feb. 2026
Cloud Computing Application Award, UCB Data Science Discovery Program	May 2024
Undergraduate Academic Excellence Scholarship	2023 – 2024
Second Prize, College Student Mathematics Competition (Hubei Division)	Mar. 2023
Second Prize, Chinese Mathematics Competition	Jan. 2023
Third Prize, China Undergraduate Mathematical Contest in Modeling	Sept. 2022